MATH329 · OPERATIONS RESEARCH

Big-M Practice Solutions

6 problems: from the slides, quiz 471, and practice

Muharib

01 Solution 1: min

Minimize $z = 2x_1 + x_2 + 0s_1 + 0s_2 + MR_1 + MR_2$	
subject to	$3x_1 + x_2 - s_1 + R_1 = 9$
	$x_1 + x_2 - s_2 + R_2 = 6$
	$x_1, x_2, s_1, s_2, R_1, R_2 \geq 0$

Min, so the Z row is fixed by adding: `New Z = Old Z + M(R1 row) + M(R2 row)`.

Basic	x_1	x_2	s_1	s_2	R_1	R_2	Sol.
R_1	3	1	−1	0	1	0	9
R_2	1	1	0	−1	0	1	6
Z	4M − 2	2M − 1	−M	−M	0	0	15M
x_1	1	$\frac{1}{3}$	$-\frac{1}{3}$	0	$\frac{1}{3}$	0	3
R_2	0	$\frac{2}{3}$	$\frac{1}{3}$	−1	$-\frac{1}{3}$	1	3
Z	0	$\frac{2M}{3} - \frac{1}{3}$	$\frac{M}{3} - \frac{2}{3}$	−M	$\frac{2}{3} - \frac{4M}{3}$	0	3M + 6
x_1	1	0	$-\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$-\frac{1}{2}$	$\frac{3}{2}$
x_2	0	1	$\frac{1}{2}$	$-\frac{3}{2}$	$-\frac{1}{2}$	$\frac{3}{2}$	$\frac{9}{2}$
Z	0	0	$-\frac{1}{2}$	$-\frac{1}{2}$	$\frac{1}{2} - M$	$\frac{1}{2} - M$	$\frac{15}{2}$

Min z = $\frac{15}{2}$ at $x_1 = \frac{3}{2}$, $x_2 = \frac{9}{2}$

02 Solution 2: max with an = constraint

Maximize $z = 2x_1 + 3x_2 - 5x_3 + 0s_2 - MR_1 - MR_2$
subject to $x_1 + x_2 + x_3 + R_1 = 7$
$2x_1 - 5x_2 + x_3 - s_2 + R_2 = 10$
$x_1, x_2, x_3, s_2, R_1, R_2 \geq 0$

Max, so we fix by subtracting: `New Z = Old Z − M(R₁ row) − M(R₂ row)`. The `=` constraint takes only `R₁`.

Basic	x_1	x_2	x_3	s_2	R_1	R_2	Sol.
R_1	1	1	1	0	1	0	7
R_2	2	−5	1	−1	0	1	10
Z	$-3M - 2$	$4M - 3$	$5 - 2M$	M	0	0	$-17M$
R_1	0	$\frac{7}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	1	$-\frac{1}{2}$	2
x_1	1	$-\frac{5}{2}$	$\frac{1}{2}$	$-\frac{1}{2}$	0	$\frac{1}{2}$	5
Z	0	$-\frac{7M}{2} - 8$	$6 - \frac{M}{2}$	$-\frac{M}{2} - 1$	0	$\frac{3M}{2} + 1$	$10 - 2M$
x_2	0	1	$\frac{1}{7}$	$\frac{1}{7}$	$\frac{2}{7}$	$-\frac{1}{7}$	$\frac{4}{7}$
x_1	1	0	$\frac{6}{7}$	$-\frac{1}{7}$	$\frac{5}{7}$	$\frac{1}{7}$	$\frac{45}{7}$
Z	0	0	$\frac{50}{7}$	$\frac{1}{7}$	$M + \frac{16}{7}$	$M - \frac{1}{7}$	$\frac{102}{7}$

$$\text{Max } z = \frac{102}{7} \quad \text{at} \quad x_1 = \frac{45}{7}, \quad x_2 = \frac{4}{7}, \quad x_3 = 0$$

03 Solution 3: quiz 471

The first constraint is `≤` and takes only `s1`, which is basic from the start. The second is `≥` and takes `−s2` and `R`.

$$\begin{aligned} &\text{Maximize } z = 2x_1 + 3x_2 - 5x_3 + 0s_1 + 0s_2 - MR \\ &\text{subject to} \quad 4x_1 + 3x_2 - x_3 + s_1 = 24 \\ &\qquad\qquad x_1 + x_2 + 2x_3 - s_2 + R = 18 \\ &\qquad\qquad x_1, x_2, x_3, s_1, s_2, R \geq 0 \end{aligned}$$

Max, so subtract `M × (R row)` from the Z row.

Basic	x_1	x_2	x_3	s_1	s_2	R	Sol.
s_1	4	3	−1	1	0	0	24
R	1	1	**2**	0	−1	1	18
Z	−M − 2	−M − 3	5 − 2M	0	M	0	−18M
s_1	$\frac{9}{2}$	**$\frac{7}{2}$**	0	1	$-\frac{1}{2}$	$\frac{1}{2}$	33
x_3	$\frac{1}{2}$	$\frac{1}{2}$	1	0	$-\frac{1}{2}$	$\frac{1}{2}$	9
Z	$-\frac{9}{2}$	$-\frac{11}{2}$	0	0	$\frac{5}{2}$	$M - \frac{5}{2}$	−45
x_2	$\frac{9}{7}$	1	0	$\frac{2}{7}$	$-\frac{1}{7}$	$\frac{1}{7}$	$\frac{66}{7}$
x_3	$-\frac{1}{7}$	0	1	$-\frac{1}{7}$	$-\frac{3}{7}$	$\frac{3}{7}$	$\frac{30}{7}$
Z	$\frac{18}{7}$	0	0	$\frac{11}{7}$	$\frac{12}{7}$	$M - \frac{12}{7}$	$\frac{48}{7}$

$$\text{Max } z = \frac{48}{7} \quad \text{at} \quad x_1 = 0,\ x_2 = \frac{66}{7},\ x_3 = \frac{30}{7}$$

04 Solution 4: min with =, ≥ and ≤

$$\begin{aligned}
&\text{Minimize } z = 4x_1 + x_2 + 0s_2 + 0s_3 + MR_1 + MR_2 \\
&\text{subject to} \quad 3x_1 + x_2 + R_1 = 3 \\
&\qquad\qquad 4x_1 + 3x_2 - s_2 + R_2 = 6 \\
&\qquad\qquad x_1 + 2x_2 + s_3 = 4 \\
&\qquad\qquad x_1, x_2, s_2, s_3, R_1, R_2 \geq 0
\end{aligned}$$

The third constraint is $\leq$ and takes only s_3, so the first tableau has R_1, R_2 and s_3. Min, so add M × the R rows.

Basic	x_1	x_2	s_2	s_3	R_1	R_2	Sol.
R_1	**3**	1	0	0	1	0	3
R_2	4	3	−1	0	0	1	6
s_3	1	2	0	1	0	0	4
Z	$7M - 4$	$4M - 1$	$-M$	0	0	0	$9M$
x_1	1	$\frac{1}{3}$	0	0	$\frac{1}{3}$	0	1
R_2	0	$\frac{5}{3}$	−1	0	$-\frac{4}{3}$	1	2
s_3	0	$\frac{5}{3}$	0	1	$-\frac{1}{3}$	0	3
Z	0	$\frac{5M}{3} + \frac{1}{3}$	$-M$	0	$\frac{4}{3} - \frac{7M}{3}$	0	$2M + 4$
x_1	1	0	$\frac{1}{5}$	0	$\frac{3}{5}$	$-\frac{1}{5}$	$\frac{3}{5}$
x_2	0	1	$-\frac{3}{5}$	0	$-\frac{4}{5}$	$\frac{3}{5}$	$\frac{6}{5}$
s_3	0	0	**1**	1	1	−1	1
Z	0	0	$\frac{1}{5}$	0	$\frac{8}{5} - M$	$-M - \frac{1}{5}$	$\frac{18}{5}$
x_1	1	0	0	$-\frac{1}{5}$	$\frac{2}{5}$	0	$\frac{2}{5}$
x_2	0	1	0	$\frac{3}{5}$	$-\frac{1}{5}$	0	$\frac{9}{5}$
s_2	0	0	1	1	1	−1	1
Z	0	0	0	$-\frac{1}{5}$	$\frac{7}{5} - M$	$-M$	$\frac{17}{5}$

- Iteration 1: x_1 enters ($7M - 4$). The ratios are 1, 1.5 and 4, so R_1 leaves.
- Iteration 2: x_2 enters. The ratios are 3, 1.2 and 1.8, so R_2 leaves.
- Iteration 3: a positive $\frac{1}{5}$ remains under s_2, so it enters. The ratios are 3 and 1, so s_3 leaves.

$$\text{Min } z = \frac{17}{5} \quad \text{at} \quad x_1 = \frac{2}{5}, \quad x_2 = \frac{9}{5}$$

Check: $3\left(\frac{2}{5}\right) + \frac{9}{5} = \frac{6}{5} + \frac{9}{5} = 3$ ✓

05 Solution 5: max with ≤ and ≥

$$\text{Maximize } z = 3x_1 + 2x_2 + 0s_1 + 0s_2 - MR$$

$$\text{subject to} \quad x_1 + x_2 + s_1 = 4$$

$$x_1 + 3x_2 - s_2 + R = 6$$

$$x_1, x_2, s_1, s_2, R \geq 0$$

Basic	x_1	x_2	s_1	s_2	R	Sol.
s_1	1	1	1	0	0	4
R	1	**3**	0	−1	1	6
Z	$-M-3$	$-3M-2$	0	M	0	$-6M$
s_1	$\frac{2}{3}$	0	1	$\frac{1}{3}$	$-\frac{1}{3}$	2
x_2	$\frac{1}{3}$	1	0	$-\frac{1}{3}$	$\frac{1}{3}$	2
Z	$-\frac{7}{3}$	0	0	$-\frac{2}{3}$	$M+\frac{2}{3}$	4
x_1	1	0	$\frac{3}{2}$	$\frac{1}{2}$	$-\frac{1}{2}$	3
x_2	0	1	$-\frac{1}{2}$	$-\frac{1}{2}$	$\frac{1}{2}$	1
Z	0	0	$\frac{7}{2}$	$\frac{1}{2}$	$M-\frac{1}{2}$	11

- Iteration 1: the most negative is `−3M − 2` under `x₂`. The ratios are `4` and `2`, so `R` leaves.

- Iteration 2: `R` has left, so `M` is gone from the columns that matter. x_1 enters ($-\frac{7}{3}$). The ratios are `3` and `6`, so s_1 leaves.

```
Max z = 11   at   x₁ = 3 ,  x₂ = 1
```

```
Check:  3 + 1 = 4 ✓    3 + 3(1) = 6 ✓
```

06 Solution 6: no feasible solution

```
Maximize z = 2x₁ + x₂ + 0s₁ + 0s₂ − MR
subject to   x₁ + x₂ + s₁ = 2
             x₁ + x₂ − s₂ + R = 4
```

Basic	x_1	x_2	s_1	s_2	R	Sol.
s_1	**1**	1	1	0	0	2
R	1	1	0	−1	1	4
Z	−M − 2	−M − 1	0	M	0	−4M
x_1	1	1	1	0	0	2
R	0	0	−1	−1	1	2
Z	0	1	M + 2	M	0	4 − 2M

After one iteration the whole Z row is ≥ 0, so the tableau is optimal. But `R` is still in the basis with value `2`, not zero.

The answer

By the slide's remark: if an artificial variable stays in the final tableau at a nonzero level, the problem has **no feasible solution**. It makes sense: no point satisfies $x_1 + x_2 \leq 2$ and $x_1 + x_2 \geq 4$ together.